

Comprehensive Summary

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Course: SIS102

Overview

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This lecture covers the topics of Insider Threat and Ransomware, which are significant concerns in the field of Systems and Internet Security Technologies. Insider threats and ransomware attacks can have devastating effects on an organization's security and data, making it essential to understand their causes, manifestations, and prevention methods. By grasping these concepts, individuals can better protect their organizations from these elusive threats.

Definitions

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- Insider threat: a threat to an organization's security that comes from within, often caused by malicious intent, mistakes, or negligence.
- Malicious insiders: individuals who intentionally harm the company.
- Negligent insiders: individuals who inadvertently cause damage through carelessness.
- Accidental insiders: individuals who fall victims to phishing or social engineering attacks.
- Ransomware: malicious software that encrypts files on a device or network, holding them hostage until a ransom is paid.
- Double extortion: a tactic used by ransomware attackers who threaten to release sensitive stolen data or report the organization to regulators if the ransom is not paid.

Main Topics

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1. Insider Threat

Insider threats are one of the most elusive threats to an organization, often caused by malicious intent, mistakes, or negligence. These threats can be manifested in different ways, including malicious insiders, negligent insiders, and accidental insiders. To prevent insider threats, organizations can implement technical controls, organizational measures, and encourage employees to report suspicious activities.

Key Points:

- Malicious intent, mistakes, and negligence are common causes of insider threats.
- Insider threats can be prevented through technical controls, organizational measures, and employee awareness.
- Encouraging employees to report suspicious activities is crucial in preventing insider threats.

Examples:

→ A disgruntled employee intentionally deleting important files is an example of a malicious insider.

→ An employee accidentally clicking on a phishing email is an example of an accidental insider.

2. Ransomware

Ransomware is a type of malicious software that encrypts files on a device or network, holding them hostage until a ransom is paid. Ransomware attackers often use double extortion tactics to increase the pressure on the organization to pay the ransom. To protect against ransomware, organizations can implement foundational cybersecurity measures, such as regular software updates and robust endpoint security solutions.

Key Points:

- Ransomware encrypts files on a device or network, holding them hostage until a ransom is paid.
- Ransomware attackers often use double extortion tactics to increase the pressure on the organization to pay the ransom.
- Foundational cybersecurity measures, such as regular software updates and robust endpoint security solutions, can help protect against ransomware.

Examples:

→ A ransomware attack that encrypts an organization's entire network is an example of a ransomware threat.

→ An organization paying a ransom to regain access to their encrypted files is an example of a response to a ransomware attack.

3. Preventing Ransomware

To prevent ransomware attacks, organizations can implement specific countermeasures, such as immutable storage architectures, application whitelisting, and network segmentation. Additionally, advanced defense tactics, such as end-to-end encryption, incident response, and threat hunting activities, can help protect against ransomware.

Key Points:

- Immutable storage architectures and application whitelisting can help prevent ransomware attacks.
- Network segmentation and advanced defense tactics, such as end-to-end encryption and incident response, can help protect against ransomware.
- Threat hunting activities and threat intelligence services can help organizations stay ahead of ransomware threats.

Examples:

- Implementing immutable storage architectures to prevent ransomware attacks is an example of a specific countermeasure.
- Conducting regular threat hunting activities to identify potential ransomware threats is an example of an advanced defense tactic.